Anukriti AI

PHARMACOGENOMICS REPORT

RESEARCH / VALIDATION

Generated: June 28, 2026
Engine: anukriti-pgx-core v0.5.0

SAMPLE ID	ANCESTRY	DATA SOURCE	SEQUENCING
HG01190	South Asian (SAS)	ENA SRR25583344	ONT Long-Read WGS

DIPLOTYPE CALLS — ALL GENES

GENE	DIPLOTYPE	PHENOTYPE	CLINICAL ACTION
CYP2D6	*68+*4x2 / *68+*4	POOR METABOLIZER	⚠ Codeine / opioid risk. Avoid or reduce TCA doses.
CYP2C19	*1 / *2	INTERMEDIATE METABOLIZER	⚡ Clopidogrel: consider alternative antiplatelet. PPIs: standard dose.
DPYD SAS	Reference / c. 85T>C (*9A)	DECREASED FUNCTION*	⚡ 5-FU / capecitabine: SAS override active. CPIC (EUR) = normal_function; Anukriti SAS label = decreased_function (Raha et al. 2025). Consider dose reduction.
CYP2C9	*1 / *61	LIMITED EVIDENCE	V3 EVIDENCEGAP *61 has limited CPIC data. Warfarin: use standard dose; monitor INR.
CYP2C8	*1 / *3	INTERMEDIATE METABOLIZER	Paclitaxel / repaglinide: monitor for reduced clearance.
CYP2B6	*1 / *5	INTERMEDIATE METABOLIZER	Efavirenz / bupropion: standard dose; consider TDM.
CYP4F2	*1 / *3 (rs2108622 het)	VARIANT CARRIER	Warfarin: may require slightly higher dose for target INR.
VKORC1	rs9923231 ref/ref (C/C)	NORMAL SENSITIVITY	Warfarin: standard dose. No sensitivity flag.
CYP1A2	*1 / *1		No action required.

GENE	DIPLOTYPE	PHENOTYPE	CLINICAL ACTION
		NORMAL METABOLIZER	
CYP3A4	*1 / *1	NORMAL METABOLIZER	No action required.
CYP3A5	*1 / *1	NORMAL METABOLIZER	Tacrolimus: no dose adjustment based on genotype alone.
G6PD	B (reference)	NORMAL ACTIVITY	Rasburicase / primaquine: no contraindication from genotype.
HLA-B	*15:20 / *15:20	NO RISK ALLELES	Not *57:01 (abacavir safe). Not *58:01 (allopurinol safe). Not *15:02 (carbamazepine — Han Chinese risk allele absent).
HLA-C	*07:01:01:128 / *07:01:01:16	CALLED	No CPIC Level A drug-gene interaction for *07:01 in current guidelines.
CFTR	Reference	NON-RESPONSIVE	Ivacaftor / lumacaftor: no responsive variant detected.

* DPYD *9A phenotype reflects Anukriti SAS population override (v5 classifier). CPIC European-reference assignment is normal_function. See §2.8 of validation paper.

KEY CLINICAL FINDINGS

CYP2D6 Poor Metabolizer — Opioid & Antidepressant Risk

Codeine and tramadol are contraindicated (CPIC Level A). Standard doses of TCAs (amitriptyline, nortriptyline) should be reduced by 25–50%. Avoid codeine entirely — PM status means negligible conversion to active morphine.

`Affected drugs: codeine, tramadol, amitriptyline, nortriptyline, paroxetine, fluoxetine, atomoxetine`

CYP2C19 Intermediate Metabolizer — Clopidogrel Reduced Efficacy

*1/*2 heterozygous. Clopidogrel (prodrug, requires CYP2C19 activation) has reduced antiplatelet effect. CPIC recommends considering prasugrel or ticagrelor for ACS patients. PPIs: standard dosing acceptable.

`Affected drugs: clopidogrel, voriconazole, amitriptyline, escitalopram, sertraline`

DPYD *9A — SAS Population-Specific Override Active

Heterozygous carrier of c.85T>C (*9A). CPIC's European-based guidelines assign normal_function. Anukriti's SAS-specific label override (based on Raha et al. 2025, South Indian colorectal cohort, ~27% SAS MAF) flags this as decreased_function. For 5-FU / capecitabine: consider 25% dose reduction and enhanced toxicity monitoring. This finding directly validates Anukriti's population-equity engine on a real SAS sample.

`Affected drugs: 5-fluorouracil, capecitabine, tegafur | Override: SAS_literature_override_CPIC_deviation`

HLA-B *15:20 — No High-Risk Alleles Detected

Neither *57:01 (abacavir hypersensitivity) nor *58:01 (allopurinol SJS/TEN) nor *15:02 (carbamazepine SJS in Han Chinese) detected. Standard use of these drugs is not contraindicated on HLA grounds.

`Cleared drugs: abacavir, allopurinol, carbamazepine (HLA-B basis)`

PIPELINE PROVENANCE

VARIANT CALLER	DIPLOTYPER	REFERENCE
Clair3 v1.0.10 (ONT WGS)	pbStarPhase 1.4.2	GRCh38 / hg38 (chr-prefix)
CPIC SNAPSHOT	**PHARMVAR**	**HLA DB**
2025-05-15	6.2.6	v3.60.0-alpha
ENGINE	**CLAIR3 MODEL**	**REPORT DATE**
anukriti-pgx-core v0.5.0	r941_prom_sup_g5014	2026-06-28